A candidate $N^{1/3+o(1)}$ upper bound for Erdős Problem 160

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This manuscript has not been peer reviewed. It proves an upper-bound improvement only and does not solve Erdős Problem 160.

Abstract

Let $H(N)$ be the least number of colours with which $\{1,\ldots,N\}$ can be coloured so that every nonconstant four-term arithmetic progression receives at least three distinct colours. We give a concrete construction with explicit palette bounds, after finite choices of a prime and a field basis, which for every integer $k \geq 1$ and

$$N > \max\{2^{(76k)^2}, 2^{(40k)^4}\},$$

satisfies

$$H(N)^{3k} < 2{,}985{,}984^k N^{k+3}.$$

Consequently $H(N) = O_\varepsilon(N^{1/3+\varepsilon})$ for every $\varepsilon > 0$, and eventually $H(N) \leq N^{1/3+\varepsilon}$. The construction first uses two subpower-size digit filters with the property that any progression receiving at most two colours from their product has the proper pattern $ABBA$. If the two $ABBA$ equalities also hold in a nine-state carry coordinate, the associated base-p digit vectors form a nonconstant affine four-term progression in $\mathbb{F}_p^3$. After an $\mathbb{F}_p$-linear identification with $\mathbb{F}_{p^3}$, the field norm for the chosen prime $p > 3$ cannot take equal values simultaneously on the outer pair and on the inner pair. Hence the combined carry-and-norm factor is $ABBA$-free and has $9p$ available colours. We also prove the exact $+1$ bridge between this least-good-colours convention and the maximum-bad-partitions convention of Erdős's original wording. The stated theorem chain is formalized in Lean 4.

1 Statement and conventions

Write $[N]_0 = \{0, 1, \ldots, N-1\}$, and likewise $[q] = \{0, 1, \ldots, q-1\}$ for a positive integer q. A nontrivial four-term arithmetic progression is an ordered quadruple

$$(x_0, x_1, x_2, x_3) = (a, a+d, a+2d, a+3d) \tag{1.1}$$

whose entries lie in $[N]_0$ and whose integer step d is nonzero. The translation $x \mapsto x+1$ identifies this convention with progressions in $\{1, \ldots, N\}$. Allowing the decreasing orientation makes no difference, since it reverses the same four-element progression. The Lean definition `siteH` uses the corresponding increasing four-element set; translation and, when necessary, reversal give the same quantified family.

Definition 1.1. For $N \geq 1$, let $H(N)$ be the least positive integer q for which there is a colouring $c : [N]_0 \to [q]$ such that

$$\bigl|\{c(x_0), c(x_1), c(x_2), c(x_3)\}\bigr| \geq 3 \tag{1.2}$$

for every progression (1.1).

The Lean definition `siteH` is total on $\mathbb{N}$; at the unused empty horizon it has the value `siteH` $0 = 0$.

The maintained Erdős Problems page uses this least-good-colours quantity [3]. Erdős's original paper instead denotes by $h(n)$ the largest number of parts for which every partition is bad: the union of two parts contains a four-term progression [1]. We denote that original extremum by $h_{\mathrm{E}}(N)$, reserving $H(N)$ for the maintained convention. For $N \geq 4$, the exact finite definition used below is

$$h_{\mathrm{E}}(N) = \max\{\, k \in \mathbb{N} : 2 \leq k \leq N \text{ and every surjective } c : [N]_0 \to [k] \text{ is bad}\,\},$$

where bad means that some nontrivial four-term progression receives at most two colours. Section 9 proves both that the set is nonempty and that this at-most-two formulation agrees with the literal distinct-nonempty-parts wording.

Theorem 1.2 (Explicit power bound)**.** *For integers $N, k \geq 1$, if*

$$N > \max\bigl\{2^{(76k)^2}, 2^{(40k)^4}\bigr\}, \tag{1.3}$$

then, with $C = 2{,}985{,}984$,

$$H(N)^{3k} < C^k N^{k+3}. \tag{1.4}$$

Corollary 1.3 (One-third upper)**.** *For every real $\varepsilon > 0$,*

$$H(N) = O_\varepsilon\bigl(N^{1/3+\varepsilon}\bigr). \tag{1.5}$$

Moreover, for every $\varepsilon > 0$, eventually

$$H(N) \leq N^{1/3+\varepsilon}. \tag{1.6}$$

For $N \geq 4$, the original convention satisfies

$$H(N) = h_{\mathrm{E}}(N) + 1, \tag{1.7}$$

including the literal interpretation in which the two parts are distinct and every displayed part is nonempty.

The maintained page currently records the exponent $\log 3/\log 22 \approx 0.3554$ [3]. The family of bounds in Corollary 1.3 has limiting exponent $1/3$, and Section 8 gives the fixed comparison $7/20 < \log 3/\log 22$. The digit-filtering framework is closely related to Deng–Tidor–Zhao [2] and to Hunter's explanation of the recorded bound [4]. A literature search did not locate a prior source for the degree-three norm factor composed with these filters, but the search was not exhaustive. We make no claim of priority or novelty.

2 The finite pattern reduction

Suppose four colours use at most two values. If no three positions have one common colour, then two distinct colours each occur exactly twice. Up to interchanging their names, the only possibilities are

$$AABB, \qquad ABAB, \qquad ABBA. \tag{2.1}$$

Thus a label excluding every three-equal pattern and $ABAB$, together with an independent label excluding $AABB$, leaves only the *proper* pattern

$$c_0 = c_3, \qquad c_1 = c_2, \qquad c_0 \neq c_1. \tag{2.2}$$

An additional $ABBA$-free label then forces at least three values on the original progression. This elementary classification is the formal theorem `atMostTwo_noThree_aabb_abab_properABBA`.

We construct three labels:

(i) one shared mod-24 label λ_N, simultaneously excluding three-equal patterns and $ABAB$;

(ii) a dyadic label μ_N, excluding $AABB$;

(iii) a cubic label ν_N, excluding $ABBA$.

Their product is therefore a good colouring. More explicitly, an at-most-two tuple of product colours projects to an at-most-two tuple in each factor. The first two labels then form a proper-$ABBA$ filter by the classification above. If adjoining the cubic label still left at most two product colours, the outer and inner equalities forced by that filter would force the two $ABBA$ equalities in the cubic coordinate, contradicting its $ABBA$-free property.

3 Fixed-length digits

For an integer base $P > 0$, put

$$Q_j(n) = \left\lfloor \frac{n}{P^j} \right\rfloor, \qquad d_j(n) = Q_j(n) \bmod P. \tag{3.1}$$

If $0 \le n < P^{L+1}$, then

$$v_P(n) = (d_0(n), \dots, d_L(n)) \in \mathbb{Z}^{L+1} \tag{3.2}$$

is a canonical fixed-length representation. In particular, $Q_L(n) < P$, so the top digit is $d_L(n) = Q_L(n)$. Equality of all coordinates in (3.2) therefore implies equality of the represented integers.

The lower digits $0, \dots, L-1$ carry auxiliary no-wrap labels. The top digit needs no auxiliary label: once the lower quotient induction has been completed, the progression relation for Q_L is already the required top digit relation. In all three digit constructions below, the square norm is

$$\|v\|^2 = \sum_{j=0}^{L} v_j^2. \tag{3.3}$$

We repeatedly use the same quotient-descent step. If a no-wrap argument shows that the current base-P residues satisfy the progression identities as integer equalities, substituting $n_j = r_j + Pq_j$, subtracting those residue identities, and dividing by P shows that the quotients q_j form the same integer progression. Iteration exposes every lower digit, and the final quotient is the top digit described above.

4 The shared mod-24 filter

Fix $B \geq 1$ and set

$$P = 24B + 1. \tag{4.1}$$

For $n < P^{L+1}$, define

$$\lambda_{P,L}(n) = \Big((d_0(n) \bmod 24, \ldots, d_{L-1}(n) \bmod 24), \|v_P(n)\|^2 \Big). \tag{4.2}$$

The available codomain, not necessarily the attained image, has exactly

$$S(P, L) = 24^L \big((L+1)P^2 + 1 \big) \tag{4.3}$$

elements. The extra endpoint in the norm coordinate is a harmless finite range containing every possible square norm.

4.1 A weighted carry-correction lemma

Lemma 4.1 (Weighted digit correction)**.** *Let α, β be positive integers with $\alpha + \beta \leq 3$, and suppose*

$$\alpha x + \beta y = (\alpha + \beta) z. \tag{4.4}$$

Write $x = r_x + Pq_x$, and similarly for y, z, with $0 \leq r_x, r_y, r_z < P$. If $r_x \equiv r_y \equiv r_z \pmod{24}$, then

$$\alpha r_x + \beta r_y = (\alpha + \beta) r_z \tag{4.5}$$

and

$$\alpha q_x + \beta q_y = (\alpha + \beta) q_z. \tag{4.6}$$

Proof. Let $E = \alpha r_x + \beta r_y - (\alpha + \beta) r_z$. Equation (4.4) gives $P \mid E$, while the residue hypothesis gives $24 \mid E$. Since $P = 24B + 1$, we have $\gcd(P, 24) = 1$, hence $24P \mid E$. On the other hand, $|E| < 3P < 24P$, so $E = 0$. Subtracting the resulting digit identity from (4.4) and dividing by P yields (4.6). □

Iterating Lemma 4.1 exposes (4.5) in every lower digit and preserves (4.4) in every quotient. The top-quotient observation in Section 3 supplies the last digit.

4.2 Excluding three equal colours

Each possible triple of positions in a four-term progression has a positive weighted relation of the form (4.4):

positions	$(x, y; z)$	(α, β)
012	$(x_0, x_2; x_1)$	$(1, 1)$
123	$(x_1, x_3; x_2)$	$(1, 1)$
013	$(x_0, x_3; x_1)$	$(2, 1)$
023	$(x_0, x_3; x_2)$	$(1, 2)$

(4.7)

If the three $\lambda_{P,L}$-colours agree, Lemma 4.1 gives digit vectors X, Y, Z satisfying

$$\alpha X + \beta Y = (\alpha + \beta) Z \tag{4.8}$$

coordinatewise, and the norm coordinates give $\|X\|^2 = \|Y\|^2 = \|Z\|^2$. For every coordinate,

$$\alpha (X_j - Z_j)^2 + \beta (Y_j - Z_j)^2 = \alpha X_j^2 + \beta Y_j^2 - (\alpha + \beta) Z_j^2. \tag{4.9}$$

After summation the right side is zero. Positivity of the weights and nonnegativity of the squares force $X = Y = Z$. Digit injectivity forces the three integers to be equal, which is impossible for three distinct positions of a nontrivial progression. Thus (4.2) excludes every three-equal pattern.

4.3 Excluding $ABAB$ with the same label

Let r_i be the current base-P residues of a four-term progression and put $\delta_i = r_{i+1} - r_i$. The progression relation before taking residues implies

$$\delta_0 = \delta_1 + uP, \qquad \delta_2 = \delta_1 + vP, \qquad u, v \in \{-1, 0, 1\}. \tag{4.10}$$

Under an $ABAB$ equality of (4.2), $r_0 \equiv r_2 \pmod{24}$ and $r_1 \equiv r_3 \pmod{24}$. Now

$$\delta_0 - \delta_1 = 2r_1 - r_0 - r_2 \tag{4.11}$$

is even, whereas P is odd. Therefore the first equality in (4.10) cannot have $u = \pm 1$, and $u = 0$. The same parity argument applied to $\delta_2 - \delta_1 = r_3 + r_1 - 2r_2$ gives $v = 0$. Iterating this no-wrap step through the quotients gives a nonconstant integer vector progression

$$X, \quad X + R, \quad X + 2R, \quad X + 3R, \qquad R \neq 0. \tag{4.12}$$

The direction R is nonzero because equality of the fixed-length digit vectors would, by Section 3, force equality of the represented integers, contrary to the original nontrivial progression.

Put $A = \langle X, R\rangle$ and $D = \|R\|^2$. The two norm equalities in the $ABAB$ pattern give

$$\begin{aligned} \|X\|^2 = \|X + 2R\|^2 \quad &\Longrightarrow \quad 4A + 4D = 0, \\ \|X + R\|^2 = \|X + 3R\|^2 \quad &\Longrightarrow \quad 4A + 8D = 0. \end{aligned} \tag{4.13}$$

Hence $D = 0$, so $R = 0$, a contradiction. The single label $\lambda_{P,L}$ therefore excludes both three-equal patterns and $ABAB$; it is counted only once in the final palette.

5 The dyadic $AABB$ filter

This construction uses the separate base B, not the mod-24 base P. For $0 \le r < B$, define the bucket

$$\beta(r) = \lfloor \log_2(r + 1) \rfloor. \tag{5.1}$$

Equality of buckets implies

$$y + 1 < 2(x + 1), \qquad x + 1 < 2(y + 1). \tag{5.2}$$

For $n < B^{L+1}$, put

$$\mu_{B,L}(n) = \Big((\beta(d_0(n)), \ldots, \beta(d_{L-1}(n))), \|v_B(n)\|^2 \Big). \tag{5.3}$$

Lemma 5.1 (Dyadic no-wrap)**.** *Suppose four residues $r_i \in [0, B)$ have congruent consecutive steps modulo B, and*

$$\beta(r_0) = \beta(r_1), \qquad \beta(r_2) = \beta(r_3). \tag{5.4}$$

Then their three signed representative steps are equal in $\mathbb{Z}$.

Proof. As in (4.10), write $\delta_0 = \delta_1 + uB$ and $\delta_2 = \delta_1 + vB$, where $u, v \in \{-1, 0, 1\}$. Equivalently,

$$2r_1 = r_0 + r_2 + uB, \qquad r_1 + r_3 = 2r_2 + vB.$$

The inequalities (5.2) give

$$r_1 \le 2r_0, \qquad r_0 \le 2r_1, \qquad r_3 \le 2r_2, \qquad r_2 \le 2r_3. \tag{5.5}$$

If $u = -1$, then $B = r_0 + r_2 - 2r_1$; since $B > r_2$, this gives $r_0 > 2r_1$, contradicting (5.5). Suppose $u = 1$. Since $B > r_1$, the first displayed identity gives $r_1 > r_0 + r_2$. If $v = -1$, then $B = 2r_2 - r_1 - r_3 > r_2$, so $r_2 > r_1 + r_3$, contradicting $r_1 > r_2$. If $v = 0$, then $r_3 = 2r_2 - r_1 \ge 0$, whence $r_2 > r_0$ and therefore $r_1 > 2r_0$, contradicting (5.5). If $v = 1$, the second displayed identity and $B > r_1$ give $r_3 > 2r_2$, again contradicting (5.5). Thus $u = 0$.

Now $2r_1 = r_0 + r_2$. If $v = -1$, then $r_3 = 2r_2 - r_1 - B < r_2 - r_1$. Together with $r_2 \le 2r_3$, this gives

$$2r_1 < r_2 \le r_0 + r_2 = 2r_1,$$

a contradiction. If $v = 1$, then $B = r_1 + r_3 - 2r_2$. The inequalities $B > r_3$ and $B > r_0 = 2r_1 - r_2$ give, respectively, $r_1 > 2r_2$ and $r_3 > r_1 + r_2$. Since $r_3 \le 2r_2$, the latter implies $r_1 < r_2$, a contradiction. Hence $v = 0$. □

Quotient induction and the top digit again turn an $AABB$ equality of $\mu_{B,L}$ into an integer vector progression (4.12). Its direction is nonzero by the same fixed-length digit injectivity argument. For $Q(t) = \|X + tR\|^2$, the two adjacent-pair norm equalities give

$$Q(0) = Q(1) \Longrightarrow 2A + D = 0, \qquad Q(2) = Q(3) \Longrightarrow 2A + 5D = 0. \tag{5.6}$$

Thus $D = 0$, contradicting $R \ne 0$. Hence (5.3) is $AABB$-free.

The available codomain has cardinality

$$D(B, L) = \big(\lfloor \log_2 B \rfloor + 1\big)^L \big((L+1)B^2 + 1\big). \tag{5.7}$$

6 All-horizon parameters and subpower bounds

Let

$$q = \operatorname{clog}_2 N, \qquad m = \lfloor \sqrt{q-1} \rfloor + 1, \qquad L = m - 1, \qquad B = 2^m, \qquad P = 24B + 1. \tag{6.1}$$

Here $\operatorname{clog}_2 N$ is the least q with $N \le 2^q$. The elementary inequality $q \le m^2$ yields

$$N \le 2^{m^2} = B^{L+1} \le P^{L+1}, \tag{6.2}$$

so both preceding digit labels cover $[N]_0$.

For the shared mod-24 factor, (4.3) becomes

$$S_N = 24^{m-1}\big(m(24 \cdot 2^m + 1)^2 + 1\big) \le 2^{8m+11}. \tag{6.3}$$

Indeed, $24^{m-1} \le 2^{5m}$, $P \le 2^{m+5}$, $m \le 2^m$, and therefore $mP^2 + 1 \le 2^{3m+11}$.

For the dyadic factor, (5.7) gives the exact available cardinality

$$D_N = (m+1)^{m-1}\big(m2^{2m} + 1\big). \tag{6.4}$$

With $r = \lceil \log_2(m+1) \rceil$, elementary estimates give

$$D_N \le 2^{mr+2m+1} \le 2^{5m\sqrt{m}}. \tag{6.5}$$

We use the following explicit thresholds. A generic palette bounded by 2^{Am+B_0} has K-th power less than N once

$$N > 2^{T^2}, \qquad T = \max\{1, 2(A+B_0)K\}. \tag{6.6}$$

To see this, put $s = \lfloor \sqrt{q-1} \rfloor$. The threshold gives $s \ge T$, while $m = s+1 \le 2s$, so $(Am+B_0)K \le s^2 \le q-1$. Applying (6.6) to (6.3) with $(A, B_0, K) = (8, 11, 2k)$ gives

$$N > 2^{(76k)^2} \quad \Longrightarrow \quad S_N^{2k} < N. \tag{6.7}$$

For (6.5), the sufficient elementary estimate used in the Lean formalization is

$$N > 2^{(20K)^4} \quad \Longrightarrow \quad D_N^K < N. \tag{6.8}$$

It follows by taking $s = \lfloor \sqrt{q-1} \rfloor$, observing that $s \ge (20K)^2$, and using $5(s+1)\sqrt{s+1}\,K \le s^2 \le q-1$. With $K = 2k$,

$$N > 2^{(40k)^4} \quad \Longrightarrow \quad D_N^{2k} < N. \tag{6.9}$$

Since $S_N, D_N > 0$, (6.7) and (6.9) imply

$$(S_N D_N)^k < N; \tag{6.10}$$

one may square both sides and use $((S_N D_N)^k)^2 = S_N^{2k} D_N^{2k} < N^2$.

7 The cubic proper-$ABBA$ factor

The preceding product leaves only the proper pattern (2.2). We now exclude it with $O(N^{1/3})$ available colours.

7.1 A nine-state carry shield

Let $p > 3$ be prime. For a residue $0 \le r < p$, define its third

$$\tau_p(r) = \begin{cases} 0, & 3r < p, \\ 1, & p \le 3r < 2p, \\ 2, & 2p \le 3r. \end{cases} \tag{7.1}$$

Write every $n < p^3$ uniquely as

$$n = r + p(w + pu), \qquad 0 \le r, w, u < p, \tag{7.2}$$

and put

$$\sigma_p(n) = (\tau_p(r), \tau_p(w)) \in \{0, 1, 2\}^2. \tag{7.3}$$

Only the two carry-bearing digits are labelled, so σ_p has nine available values.

Lemma 7.1 (Thirds no-wrap)**.** *Let $r_0, r_1, r_2, r_3 \in [0, p)$, with their consecutive signed differences congruent modulo p. If*

$$\tau_p(r_0) = \tau_p(r_3), \qquad \tau_p(r_1) = \tau_p(r_2), \tag{7.4}$$

then the three representative differences are equal in $\mathbb{Z}$.

Proof. Let the common outer region in (7.4) be a, and the common inner region be b. Write

$$3r_j = \rho_j p + e_j, \qquad 0 \le e_j < p, \qquad \rho_0 = \rho_3 = a, \qquad \rho_1 = \rho_2 = b.$$

Every representative difference lies strictly between $-p$ and p. Consequently, relative to the middle difference, there are $u, v \in \{-1, 0, 1\}$ such that

$$\delta_0 = \delta_1 + up, \qquad \delta_2 = \delta_1 + vp.$$

Put $d = b - a \in \{-2, -1, 0, 1, 2\}$. Multiplying the two difference identities by 3 and using the displayed decompositions gives

$$(d - 3u)p = e_0 + e_2 - 2e_1, \qquad (-d - 3v)p = 2e_2 - e_1 - e_3. \tag{7.5}$$

Both right-hand sides lie strictly between $-2p$ and $2p$. Since the coefficients on the left are integers, the only possibilities are

d	u	v
-2	-1	1
-1	0	0
0	0	0
1	0	0
2	1	-1

When $d = -2$, subtracting the second equation in (7.5) from the first would give

$$2p = e_0 + e_3 - e_1 - e_2 < 2p,$$

which is impossible. When $d = 2$, subtracting the first equation from the second similarly gives

$$2p = e_1 + e_2 - e_0 - e_3 < 2p.$$

Thus the two extreme rows are impossible, and all remaining rows have $u = v = 0$, as required. This argument is formalized as `thirdsRegion_noWrap`. □

Apply Lemma 7.1 first to the low digits in (7.2). To spell out the quotient descent, write $n_j = r_j + pq_j$. The original progression identities and the exact residue identities give

$$p(q_0 + q_2 - 2q_1) = 0, \qquad p(q_1 + q_3 - 2q_2) = 0.$$

Since $p \neq 0$, the quotients $q_j = s_j + pu_j$ form an integer progression. Apply the lemma again to the middle digits; the same division then leaves an integer progression in the high digits. Thus, whenever a nontrivial four-term progression has $ABBA$ equality under the shield (7.3), its integer digit vectors form an affine progression. Reducing the three coordinates modulo p gives

$$X, \quad X + R, \quad X + 2R, \quad X + 3R \tag{7.6}$$

over $\mathbb{F}_p^3$, with $R \neq 0$. The shield does not make arbitrary progressions carry-free; this conclusion specifically uses the two $ABBA$ shield equalities. For completeness, R remains nonzero after reduction: the integer digit step is nonzero by injectivity of the fixed-length representation, every coordinate has absolute value less than p, and so coordinatewise divisibility by p would force that step to be zero.

7.2 The degree-three norm

Let

$$\overline{D}_p(n) = (r \bmod p, w \bmod p, u \bmod p) \in \mathbb{F}_p^3$$

be the reduced digit vector from (7.2). Choose an $\mathbb{F}_p$-linear isomorphism $\iota_p : \mathbb{F}_p^3 \xrightarrow{\sim} \mathbb{F}_{p^3}$, and let

$$\mathcal{N} : \mathbb{F}_{p^3} \longrightarrow \mathbb{F}_p \tag{7.7}$$

be the field norm. On every affine line it is a cubic polynomial:

$$\mathcal{N}(X + tR) = at^3 + bt^2 + ct + d, \qquad a = \mathcal{N}(R). \tag{7.8}$$

This follows, for example, by writing the norm as the determinant of the 3×3 matrix of multiplication and taking the leading determinant.

Apply ι_p to (7.6), and denote the resulting base point and nonzero direction again by X, R. Suppose their four norm values have the $ABBA$ pattern. From $f(0) = f(3)$ and $f(1) = f(2)$, where the right side of (7.8) is $f(t)$, we obtain

$$27a + 9b + 3c = 0, \qquad 7a + 3b + c = 0. \tag{7.9}$$

Subtracting three times the second equation from the first gives $6a = 0$. Because $p > 3$, the element 6 is nonzero in $\mathbb{F}_p$, so $\mathcal{N}(R) = a = 0$. A field norm vanishes only at zero, contradicting $R \neq 0$.

It follows that

$$\nu_p(n) = \big(\sigma_p(n), \mathcal{N}\big(\iota_p(\overline{D}_p(n))\big)\big) \tag{7.10}$$

is $ABBA$-free on $[0, p^3)$, with exactly $9p$ available codomain values.

7.3 A prime for every horizon and the constant

Let $q = \operatorname{clog}_2 N$, and define

$$e = \left\lfloor \frac{q+2}{3} \right\rfloor + 2, \qquad y = 2^e. \tag{7.11}$$

Then $y > 3$ and $N \leq y^3$. Bertrand's postulate supplies a prime p with

$$y < p \leq 2y, \tag{7.12}$$

so $N \leq p^3$ and (7.10) covers $[N]_0$. If $N > 1$, then $3e \leq q + 8$ and $2^{q-1} < N$. Consequently

$$y^3 \leq 2^{q+8} < 512N, \qquad p^3 < 4096N. \tag{7.13}$$

Therefore the cubic available palette $U_N = 9p$ satisfies

$$U_N^3 = (9p)^3 < 729 \cdot 4096\, N = 2{,}985{,}984N. \tag{7.14}$$

No optimization of this uniform constant is intended.

8 Proof of the main bound

Take the product colouring

$$c_N(n) = \bigl(\lambda_N(n), \mu_N(n), \nu_N(n)\bigr). \tag{8.1}$$

The shared label excludes three-equal patterns and $ABAB$, the dyadic label excludes $AABB$, and the cubic label excludes the remaining proper $ABBA$. Section 2 therefore shows that every nontrivial four-term progression receives at least three product colours. Hence

$$H(N) \le S_N D_N U_N. \tag{8.2}$$

Assume (1.3). Equations (6.10) and (7.14) give

$$\begin{aligned} H(N)^{3k} &\le (S_N D_N)^{3k} U_N^{3k} \\ &< N^3 \bigl(2{,}985{,}984 N\bigr)^k \\ &= 2{,}985{,}984^k N^{k+3}, \end{aligned} \tag{8.3}$$

which proves Theorem 1.2.

Taking positive $3k$-th roots gives

$$H(N) < C^{1/3} N^{1/3+1/k}. \tag{8.4}$$

Given $\varepsilon > 0$, choose k with $1/k < \varepsilon$. Equation (8.4) first yields (1.5). For (1.6), put $\delta = \varepsilon - 1/k > 0$; eventually $C^{1/3} \le N^{\delta}$, so the fixed coefficient is absorbed into the remaining exponent.

As a completely rational visible comparison, $k = 60$ gives exponent $(60+3)/(180) = 7/20$. The integer inequality

$$22^7 < 3^{20} \tag{8.5}$$

implies $7/20 < \log 3/\log 22$, so even this fixed specialization improves the exponent recorded on the maintained page.

9 The original partition convention

We now justify (1.7) rather than silently identifying two different definitions. Fix $N \ge 4$ and write $H = H(N)$. First, $H \le N$: the injective N-colouring is good because every nontrivial progression has four distinct entries. Choose a good H-palette colouring and compress its codomain to its image, of size $\ell \le H$. The compressed colouring is surjective and good. Minimality of H forces $\ell = H$, so a surjective good H-colouring exists.

If a surjective good colouring has $k < N$ colours, one fibre has at least two elements. Splitting one element into a fresh colour preserves goodness. Iteration therefore gives a surjective good colouring for every $H \le k \le N$. Conversely, if $k < H$, minimality of H says that no k-colouring is good, so in particular every surjective k-colouring is bad.

No two-colouring is good because the progression $0, 1, 2, 3$ itself uses at most two colours. Thus $H \ge 3$, the finite set defining $h_{\mathrm{E}}(N)$ is nonempty, and the preceding two paragraphs show that its maximum is exactly $H - 1 \ge 2$.

Finally, when $2 \le k \le N$, the literal phrase "the union of two distinct nonempty parts contains a progression" is equivalent to saying that some progression uses at most two colours. If it uses two colours, choose those two parts. If it uses only one, choose that part and any other nonempty part. The converse is immediate. This proves

$$h_{\mathrm{E}}(N) = H(N) - 1 \qquad (N \ge 4), \tag{9.1}$$

under both readings, and hence (1.7). The corresponding original-convention version of (1.4) is therefore

$$(h_{\mathrm{E}}(N)+1)^{3k} < C^k N^{k+3}. \tag{9.2}$$

10 Lean formalization

The Lean development uses Lean 4.30.0 and mathlib 4.30.0, pinned by the repository manifest. The principal theorem names are listed below.

Mathematical role	Lean theorem
two-colour classification	`atMostTwo_noThree_aabb_abab_properABBA`
shared mod-24 filter	`horizonNoThreeMod24Colour_isNoThreeEqualColouring`, `horizonABABMod24Colour_isABABFreeColouring`
dyadic $AABB$ filter	`horizonDyadicSquareColour_isAABBFreeColouring`
cubic $ABBA$ factor	`horizonCubicColour_abba_free`
displayed threshold bridge	`explicitUnifiedOneThirdThreshold_eq_displayed`
exact natural-power bound	`siteH_oneThirdPlusSubpower_unified_displayed`
real epsilon Big-O	`siteH_oneThirdPlusEpsilon_unified_isBigO`
coefficient-one eventual bound	`siteH_le_rpow_oneThird_add_epsilon_eventually`
strict source bridge	`siteH_eq_sourceSurjectiveOriginalH_add_one`
literal distinct-two characterization	`sourceSurjectiveOriginalH_literal_distinct_two_characterization`

(10.1)

The repository check builds the Lean project, scans the E160 sources for `sorry` and `admit`, prints the axioms of the listed theorems, and verifies the hashes of short source excerpts. The reported axioms are standard Lean/mathlib principles such as propositional extensionality, classical choice, and quotient soundness; no project-defined axiom or `sorryAx` is used.

Kernel checking establishes the formal implications represented by the code; source interpretation, novelty, and peer review lie outside that check. No asymptotic colouring theorem from [2] is imported: the mod-24 and dyadic constructions used above are proved explicitly in the Lean development.

11 LLM assistance

LLMs accessed through ChatGPT Pro and OpenAI Codex, primarily GPT-5.6, were used during proof exploration, Lean formalization, testing, and editing. The author is responsible for the manuscript and code; LLM outputs were not treated as proof or independent review.

References

[1] P. Erdős, *Some problems and results on combinatorial number theory*, in Graph Theory and Its Applications: East and West, Annals of the New York Academy of Sciences 576 (1989), 132–145. Problem passage on printed p. 143.

[2] M. Deng, J. Tidor, and Y. Zhao, *Uniform sets with few progressions via colorings*, Mathematical Proceedings of the Cambridge Philosophical Society 179 (2025), 79–103; arXiv:2307.06914.

[3] T. Bloom, ed., *Erdős Problem 160*, maintained problem page, `https://www.erdosproblems.com/160`, accessed 14 July 2026.

[4] Z. Hunter, comment on Erdős Problem 160, `https://www.erdosproblems.com/forum/thread/160`, posted 18 October 2025 (displayed time has no stated timezone), accessed 14 July 2026.

[5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Computer Science 12699 (2021), 625–635.